

ANALYSTLAB AFRICA | INTERNSHIP PROGRAMME

3 FINAL SUBMISSION: ADVANCED BUSINESS INTELLIGENCE & VALIDATION REPORT

Project Title: FinTrust Financial Intelligence & Digital Banking Analytics

Track: Data Analytics Track

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Role / Track: Data Analytics

Deliverables Included: PART E (Validation of Business Findings) & PART F (Management Recommendations)

PART E: VALIDATION OF BUSINESS FINDINGS

Finding 1: Dominance of Mobile App as the Primary Transaction Channel

Finding: Mobile App serves as FinTrust's primary digital transaction channel, accounting for the highest overall transaction volume.

Evidence: The "Transaction Volume By Channel" visual in Power BI shows Mobile App leading with 5.1K transactions, compared to POS (2.4K), Web (1.5K), ATM (1.7K), and USSD (0.5K).

Validation: Verified via SQL aggregation (`COUNT(Transaction_ID) GROUP BY Channel`) and Python Pandas value counts, confirming an exact match with the Power BI visual metrics.

Business Meaning: Mobile user experience, server uptime, and app speed are the most critical factors driving customer activity and bank revenue.

Finding 2: Flagged Risk Review Rate

Finding: Approximately 19.60% of all processed transactions trigger a risk review flag.

Evidence: The "Flagged Risk Rate %" KPI Card displays 19.60% across a total of 12K logged transactions.

Validation: Cross-checked by running `AVG(Risk_Flag) * 100` in SQL and validating in Python, returning 19.60%.

Business Meaning: Nearly 1 in every 5 transactions undergoes risk review, highlighting significant operational demand on compliance review teams.

Finding 3: Uniform Risk Flag Distribution Across Customer Segments

Finding: Transaction risk flags are evenly distributed across all customer segments rather than concentrated in a single high-risk group.

Evidence: The "Risk Breakdown By Customer Segment" donut chart reveals virtually identical proportions: Premium (25.81%), Student (25.07%), SME

(24.63%), and Everyday (24.49%).

Validation: Grouped aggregation queries in SQL and Python confirm that risk flags occur at consistent rates across all account tiers.

Business Meaning: Security rules and fraud monitoring cannot be limited to specific customer tiers; risk control mechanisms must be applied universally.

Finding 4: Settlement Health and Transaction Status Breakdown

Finding: Successful transactions represent the majority of system activity, while failed and reversed transactions account for smaller friction points.

Evidence: The "Transaction Status By Breakdown" visual displays over 10.9K successful transactions, alongside lower volumes of failed (0.4K), reversed (0.3K), and pending (0.1K) transactions.

Validation: Verified using SQL GROUP BY Transaction_Status and cross-referenced with Python frequency distributions.

Business Meaning: Core payment clearing is reliable, but failed/reversed transactions represent customer experience friction that requires automated resolution.

Finding 5: Average Transaction Size and Total Processed Volume

Finding: FinTrust processed ₦560.48M in total transaction volume, with an average transaction value of ₦46.71K.

Evidence: KPI Cards display "Total Volume NGN" as ₦560.48M and "Avg Transaction Value NGN" as ₦46.71K.

Validation: Calculated using SUM(Amount_NGN) and AVG(Amount_NGN) in SQL and Python, matching DAX card measures.

Business Meaning: Substantial financial liquidity flows through the platform, underscoring the economic necessity of robust automated fraud prevention.

PART F: MANAGEMENT RECOMMENDATIONS

Recommendation 1: Prioritize Mobile App Infrastructure and Performance

Finding: Mobile App drives over 40% of overall transaction volume (5.1K out of 11.2K channel transactions).

Evidence: Mobile App volume (5.1K) significantly exceeds all other channels.

Business Implication: System downtime or degraded mobile performance immediately restricts transaction throughput and risks customer churn.

Recommended Action: Allocate engineering resources to optimize mobile API latency, conduct load testing for high traffic periods, and schedule maintenance strictly during off-peak hours.

Recommendation 2: Refine Automated Risk Detection Rules

Finding: The overall transaction risk review flag rate stands at 19.60%.

Evidence: "Flagged Risk Rate %" KPI card shows 19.60% across 12K transactions.

Business Implication: A high flag rate generates false positives, increasing manual compliance workload and delaying legitimate payments.

Recommended Action: Fine-tune fraud detection thresholds in collaboration with compliance teams to filter out low-risk routine transactions while maintaining strict oversight on high-value transfers.

Recommendation 3: Implement Universal Multi-Factor Security Rules

Finding: Risk flag rates are balanced across Premium (25.81%), Student (25.07%), SME (24.63%), and Everyday (24.49%) tiers.

Evidence: Donut chart breakdown shows less than 1.5% variance across all customer segments.

Business Implication: Fraud risks affect high-tier and low-tier accounts equally.

Recommended Action: Enforce mandatory Multi-Factor Authentication (MFA) and adaptive behavioral security protocols uniformly across all customer account tiers.

Recommendation 4: Automate Failed Transaction Reversals

Finding: Non-successful transactions (failed, reversed, pending) account for over 800 logged events.

Evidence: Transaction Status visual highlights non-zero counts for failed (0.4K) and reversed (0.3K) transactions.

Business Implication: Settlement delays increase customer support query volume and diminish user trust.

Recommended Action: Deploy automated instant-reversal triggers for network timeouts and implement auto-retry routing across payment gateways.

Recommendation 5: Expand Adoption Strategies for Low-Volume Channels

Finding: USSD records the lowest transaction volume across all channels (0.5K transactions).

Evidence: USSD volume (0.5K) represents less than 10% of Mobile App volume (5.1K).

Business Implication: Offline and feature-phone users are underutilizing FinTrust banking services.

Recommended Action: Simplify USSD menu flows, lower USSD session fee structures, and launch targeted awareness campaigns for rural and offline banking demographics.